

EE 503: Advance Communication Theory

Course Code : EE 503

Course Name : Advance Communication Theory

L-T-P-C : 3-0-0-3

Prerequisites : IC-210: Probability, Statistics and Random Processes, EE-304: Communication Theory and the instructors consent

Intended for :UG/MS/PhD

Distribution : Elective for EE

Approval: 5th Senate

Course Contents

- **Module I:** Review of digital modulation schemes for baseband and bandlimited channels and their corresponding optimal detectors and error probabilities [6 Lectures]
- **Carrier and Symbol Synchronization:** importance in signal demodulation, carrier frequency and phase estimation decision directed and power of N methods, timing estimation - spectral-line, MMSE, and ML methods, joint carrier and symbol synchronization. [6 Lectures]
- **Equalization:** Optimal zero-forcing equalization, Linear, Decision-feedback, Adaptive Linear, Adaptive Decision-feedback, and Blind equalization. [6 Lectures]
- **Multichannel and Multicarrier Systems:** AWGN multichannels, Multicarrier communications: OFDM modulation and demodulation, spectral characteristics, bit and power allocation, channel coding. [6 Lectures]
- **Spread Spectrum Communications:** model, Direct sequence SS, PN sequences, Frequency-hopped SS, synchronization, jamming, CDMA. [5 Lectures]
- **Introduction to Information and Coding Theories:**
 - Information Theory: information measures, Shannon entropy, differential entropy, mutual information, capacity theorem for point-to-point channels with discrete and continuous alphabets. [4 Lectures]
 - Coding Theory: linear block codes definitions, properties, bounds on minimum distance (singleton, Hamming, GV, MRRW), soft versus hard decision decoding, some specific codes (Hamming, RS, Concatenated); Convolutional codes structure, decoding (the Viterbi and BCJR algorithms); Turbo codes, LDPC codes. [9 Lectures]

Reference Books:

1. T. Cover and J. Thomas, **Elements of Information Theory**, 2nd Edition, Wiley, 2006.
2. R. G. Gallager, **Principles of Digital Communication**, Cambridge Univ. Press, 2008.
3. A. Lapidoth, **A Foundation in Digital Communication**, Cambridge Univ. Press, 2009.
4. S. Lin and D. Costello, **Error Control Coding**, 2nd Edition, Prentice Hall, 2004.



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5. J. G. Proakis and M. Salehi, **Digital Communications**, 5th Edition, McGraw-Hill, Prentice Hall, 2007.
 6. B. Sklar, **Digital Communications: Fundamentals and Applications**, 2nd Edition, Prentice Hall, 2001.